MATHEMATICS

GENERAL NOTATION, SETS AND PROBABILITY

General notation

Symbol	Meaning	Examples
$=,\ \neq$	Equal; not equal	$2+3=5;\quad 2\neq 3$
$\approx$	Approximately equal	$\pi\approx 3.14;\quad \sqrt{2}\approx 1.41$
$<,\ >$ $\leq,\ \geq$	Less than; greater than Less/greater than or equal to	$2<3;\quad 5>4$ $2\leq 2;\quad 5\geq 4$
$\pm$	Plus or minus: two choices	$3\pm 1=4$ or $2;\quad -2\pm 3=1$ or -5
$\lvert a\rvert$	Absolute value: distance from a to 0	$\lvert -5\rvert=5;\quad \lvert 3\rvert=3;\quad \lvert 0\rvert=0$
$\sqrt{a}$	Nonnegative square root; for real values, $a\geq 0$	$\sqrt{9}=3$ (not ± 3); $\quad\sqrt{\frac{1}{4}}=\frac{1}{2}$
$ab,\ a\cdot b,\ a^n$	Multiplication; a^n: n factors of a (n a positive integer)	$2x=2\cdot x;\quad 3^2=3\cdot 3=9$

Sets and number systems

Symbol	Meaning	Examples
$\{a,b,c\}$	Set: order and repetition do not matter.	$\{2,4\}=\{4,2\}$ $\{2,2,4\}=\{2,4\}$
$\in$ $\notin$	Is an element of Is not an element of	$2\in\{1,2\}$ $3\notin\{1,2\}$
$\{x\in A : p(x)\}$	Set-builder: x in A such that $p(x)$ holds. : or | means "such that".	$\{x\in\mathbb{R} : x>0\}$ Positive real numbers
$\varnothing,\ \lvert A\rvert$	Empty set: no elements. $\lvert A\rvert$ counts a finite set's elements.	$\lvert\varnothing\rvert=0$ $\lvert\{2,4,6\}\rvert=3$
$A\subseteq B$	Subset: all elements of A are in B; may be equal.	$\{1\}\subseteq\{1,2\}$ $\{1,2\}\subseteq\{1,2\}$
$A\subsetneq B$	Proper subset: $A\subseteq B$ and $A\neq B$.	$\{1\}\subsetneq\{1,2\}$
$A\cup B,\ A\cap B$	Union: in either or both. Intersection: in both.	$\{1,2\}\cup\{2,3\}=\{1,2,3\}$ $\{1,2\}\cap\{2,3\}=\{2\}$
$A\setminus B$	Set difference: in A but not in B.	$\{1,2\}\setminus\{2,3\}=\{1\}$
$\mathbb{N},\ \mathbb{Z}$	Natural numbers; integers (including 0 and negatives).	Here $\mathbb{N}=\{1,2,3,\ldots\}$. Some texts include 0; check. $-2,0,3\in\mathbb{Z}$
$\mathbb{Q},\ \mathbb{R}$	Rationals: integer ratios p/q, $q\neq 0$. Reals: the number line.	$\frac{1}{2},-3\in\mathbb{Q}$ $\sqrt{2},\pi\in\mathbb{R}$
$\mathbb{R}\setminus\mathbb{Q}$	Irrationals: reals that are not rational.	$\sqrt{2},\pi\in\mathbb{R}\setminus\mathbb{Q}$
$\mathbb{N}\subsetneq\mathbb{Z}$	Number systems nest.	$\mathbb{N}\subsetneq\mathbb{Z}\subsetneq\mathbb{Q}\subsetneq\mathbb{R}$
$(a,b),\ [a,b]$	Intervals ($a<b$): parentheses exclude endpoints; brackets include them.	$x\in(1,3)$: $1<x<3$ $x\in[1,3]$: $1\leq x\leq 3$
$[a,b),\ (a,b]$	Half-open intervals: include just one endpoint.	$2\in[2,5),\ 5\notin[2,5)$ $2\notin(2,5],\ 5\in(2,5]$
$\infty,\ -\infty$	Unbounded directions; not reals. Use parentheses at infinity.	$[0,\infty)=\{x\in\mathbb{R} : x\geq 0\}$

Dots in lists: see Poster 2. Bars mean absolute value for numbers and cardinality for finite sets.

Probability and events

Example contexts
Fair die: $\Omega=\{1,2,3,4,5,6\}$.
$A=\{1,3,5\}$ (odd), $B=\{1,2\}$.
Unless stated otherwise, examples below use this die.

Symbol	Meaning	Examples
$\Omega,\ \omega$	Sample space Ω: all possible outcomes. Outcome ω: one result.	$\omega=3\in\Omega$
$A\subseteq\Omega$	An event is a set of outcomes from the sample space.	$A=\{1,3,5\}$: odd result
$P(A)$	Probability of A; always between 0 and 1.	$0\leq P(A)\leq 1$ $P(\Omega)=1,\ P(\varnothing)=0$
$\frac{\lvert A\rvert}{\lvert\Omega\rvert}$	Equals $P(A)$ only for a finite sample space of equally likely outcomes.	$P(A)=\frac{3}{6}=\frac{1}{2}$
A^c	Complement: outcomes in Ω outside A.	$A^c=\{2,4,6\}$ $P(A^c)=1-P(A)=\frac{1}{2}$
$A\cup B,\ A\cap B$	Apply set operations (left): A or B (or both); A and B.	$A\cup B=\{1,2,3,5\}$ $A\cap B=\{1\}$
$A\cap C=\varnothing$	Mutually exclusive events cannot occur together.	$C=\{2,4,6\}$ $A\cap C=\varnothing$
$P(A\mid B)$	Conditional probability: chance of A given B; requires $P(B)>0$.	$P(A\mid B)=\frac{P(A\cap B)}{P(B)}$ $=\frac{1/6}{2/6}=\frac{1}{2}$
$P(A\cap B)$	For independent events, $P(A\cap B)=P(A)P(B)$. If $P(B)>0$, this means $P(A\mid B)=P(A)$.	New example: two independent fair coin tosses. E: first is heads; F: second is heads. $P(E\cap F)=\frac{1}{4}=\frac{1}{2}\cdot\frac{1}{2}$

Keep the ideas separate.
Mutually exclusive events of positive probability are not independent: knowing one occurred rules out the other.

The bar in $P(A\mid B)$ means "given"; the bar in set-builder notation means "such that". Context determines the reading.

MATHEMATICS

FUNCTIONS, VARIABLES, SEQUENCES AND SUMS

Function notation

Example contexts
Example: $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^2$.
Set and interval notation: see Poster 1.

Symbol	Meaning	Examples
f	A function assigns exactly one output to each input in its domain. f is its name.	$f(x) = x^2$
$f(x)$	Value at input x; parentheses mark the argument, not f multiplied by x.	$f(3) = 3^2 = 9$
$f : A \to B$	Domain A: allowed inputs. Codomain B: declared target set for outputs.	$f : \mathbb{R} \to \mathbb{R}$ Domain and codomain: $\mathbb{R}$
$x \mapsto f(x)$	Input-to-output rule; $\to$ above connects sets, $\mapsto$ connects values.	$x \mapsto x^2$; $\quad 3 \mapsto 9$
$\mathrm{dom}(f)$	Domain notation; inputs must belong to this set.	$\mathrm{dom}(f) = \mathbb{R}$
$\mathrm{range}(f)$	Range: outputs actually attained; a subset of the codomain.	$\mathrm{range}(f) = [0, \infty)$ Negative reals are not outputs.

Letters and Greek symbols

Symbol	Name / description
$a, b, c;\ x, y, z$	Common lowercase Latin letters
A, B, C	Common uppercase Latin letters
$\alpha,\ \beta,\ \gamma$	Alpha; beta; gamma
$\delta,\ \Delta$	Delta; uppercase delta
$\theta,\ \Theta$	Theta; uppercase theta
$\lambda,\ \Lambda$	Lambda; uppercase lambda
$\mu,\ \sigma,\ \Sigma$	Mu; sigma; uppercase sigma
$\omega,\ \Omega$	Omega; uppercase omega (Poster 1)
π	Pi; in these examples, the circle constant $\pi \approx 3.14$.

Names do not fix meanings. Σ is a letter; $\sum$ is the related summation operator (right), not an ordinary variable.

Variables

A **variable** represents values allowed by its context. A **constant** stays fixed within that context. A letter can serve either role.

Separate choices, not simultaneous equalities

x might be -2, 0, or 3.5 in separate cases.
$n \in \mathbb{N}$ may count objects; $\theta = 30^\circ$ may name an angle ($^\circ$ means degrees).
A may be a set such as $\{1, 2\}$; λ may be a number such as 2.

State what is fixed and what varies

In $y = mx + b$, x is the input and y the output; m and b are fixed for a chosen line.

Uppercase letters often name sets; Greek letters often name angles or numbers. These are conventions, not permanent types.

Lists, sequences and subscripts

A **sequence** is an ordered list of terms. A **series** is a sum of sequence terms. Here, parentheses show sequences; braces show sets (Poster 1).

Dots omit entries in an established pattern

… in lists and ⋯ in sums indicate omitted entries following an established pattern. The dots alone do not define a unique rule.

$1, 2, 3, \ldots, n$ ends at n; $\quad n \geq 3$.
$1, 2, 3, \ldots$ continues indefinitely.

A final term marks a finite list; without a final term, the second example continues without end.

A subscript identifies a term

In a_i, the **index** i identifies a position or label; it is not multiplication or an exponent.

a_1 = term at index 1,
a_n = term at index n.

Indices often start at 1, but can start at 0 or another integer: state the starting point. Subscripts may also label cases: A_1, A_2 can name alternatives.

Finite and infinite sequences

For the rule $a_i = 2i$:
$(a_i)_{i=1}^{4} = (a_1, a_2, a_3, a_4) = (2, 4, 6, 8)$,
$(a_i)_{i=1}^{\infty} = (2, 4, 6, 8, \ldots)$.

The first has four terms; the second has no last term. With this rule, $a_3 = 2(3) = 6$.

Order and repetitions matter

$(2, 4) \neq (4, 2)$ and $(2, 2, 4) \neq (2, 4)$. Sets behave differently: see Poster 1.

Summation notation

$$\sum_{i=1}^{n} a_i$$

$\sum$: sum the indicated terms
i: index; 1: lower bound
n: upper bound
a_i: summand (expression summed)

The index runs through consecutive integers from the lower bound to the upper bound, **inclusive**. Here n is a positive integer.

$\sum_{i=1}^{n} a_i = a_1 + a_2 + \cdots + a_n$.

The centered dots follow the same omission convention as the list dots above.

One term versus the sum of the first n terms

Define S_n as the sum of the first n terms (indexing from 1):

$S_n = \sum_{i=1}^{n} a_i$.
$a_i = 2i$: $\quad a_4 = 8, \qquad S_4 = 2 + 4 + 6 + 8 = 20$.

Other summands and bounds

Constant: $\quad \sum_{i=2}^{4} 3 = 3 + 3 + 3 = 9$.

Squares: $\quad \sum_{k=1}^{3} k^2 = 1 + 4 + 9 = 14$.

The index name can change

$\sum_{i=1}^{4} i = \sum_{k=1}^{4} k = 1 + 2 + 3 + 4 = 10$.

Change the index consistently in both the bounds and the summand. The index is local to its sum.